

Mecha Connect AI

Intelligent Roadside Assistance using Machine Learning & Generative AI

Abstract

Vehicle breakdowns are unpredictable and stressful, largely because owners lack immediate access to reliable diagnostic guidance, verified repair professionals, and transparent cost information. **Mecha Connect AI** addresses this gap through a full-stack mobile application that pairs a conventional service-marketplace workflow with a layered artificial intelligence pipeline. The system is built on a Flutter front end for vehicle owners and mechanics, a Python backend that orchestrates all business logic, and a Firebase / MySQL data layer that manages users, vehicles, service requests, and mechanic profiles.

On top of this foundation, the platform applies Machine Learning (Scikit-learn) to translate user-reported symptoms into a ranked list of probable faults, an associated confidence score, and an estimated repair cost range; the same pipeline scores and ranks nearby verified mechanics using location, specialization, and rating data retrieved through the Google Maps API. Natural Language Processing is used to clean and structure free-text symptom descriptions before they reach the prediction models. A Generative AI assistant, grounded through Retrieval-Augmented Generation (RAG) over vehicle manuals and FAQ content, answers maintenance questions with source-backed responses rather than relying on the language model alone. Lightweight AI agents automate downstream operational tasks — booking the recommended mechanic, tracking the status of a service request, and issuing preventive-maintenance reminders — closing the loop between diagnosis and resolution without manual follow-up. Deep Learning-based image damage detection is scoped as a near-term extension once labelled vehicle-damage image data is available.

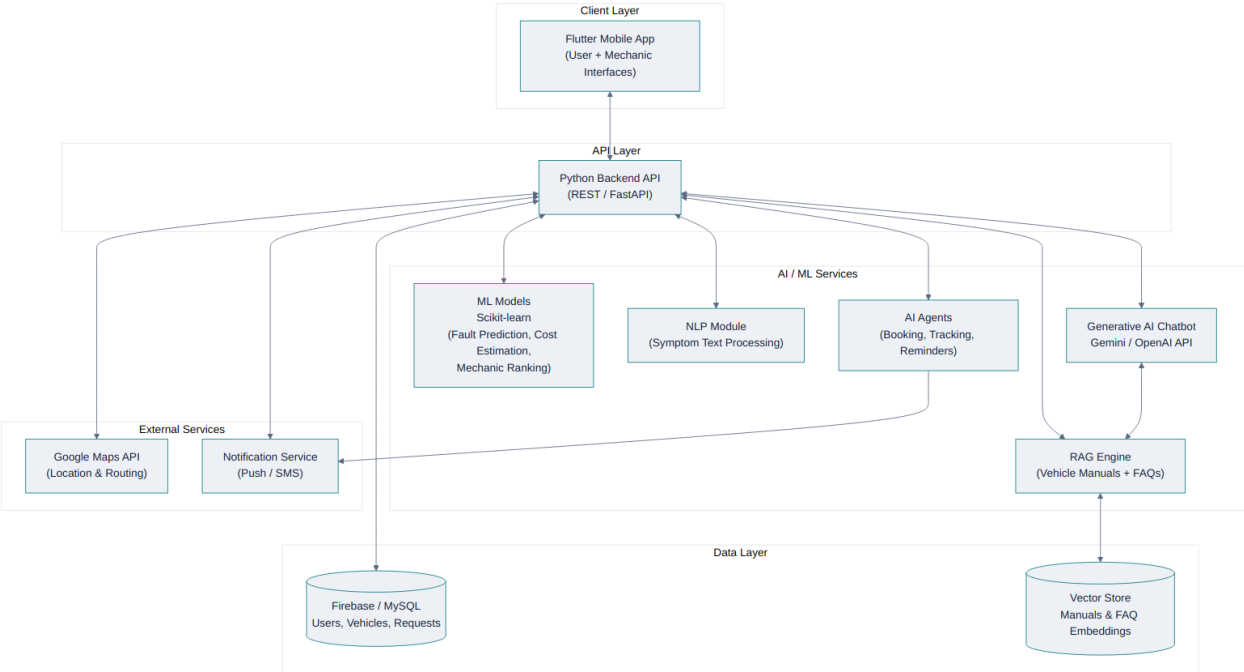
Beyond the AI layer, the project models the application as a complete system: a layered architecture spanning client, API, AI-services, data, and external-integration tiers; a use-case model capturing the distinct responsibilities of vehicle owners, mechanics, and administrators; an entity-relationship model for the underlying relational schema; a data-flow model tracing how symptom, prediction, booking, and chat information move through the system; and a sequence model illustrating the end-to-end booking interaction. Together, these models demonstrate that Mecha Connect AI is not an isolated AI prototype but an engineered, end-to-end product: one that is expected to shorten the time between breakdown and repair, improve the accuracy of fault diagnosis, and give users cost transparency and peace of mind during an otherwise stressful situation.

Keywords: Roadside Assistance, Machine Learning, Natural Language Processing, Generative AI, Retrieval-Augmented Generation, AI Agents, Mobile Application, Predictive Maintenance

System Models

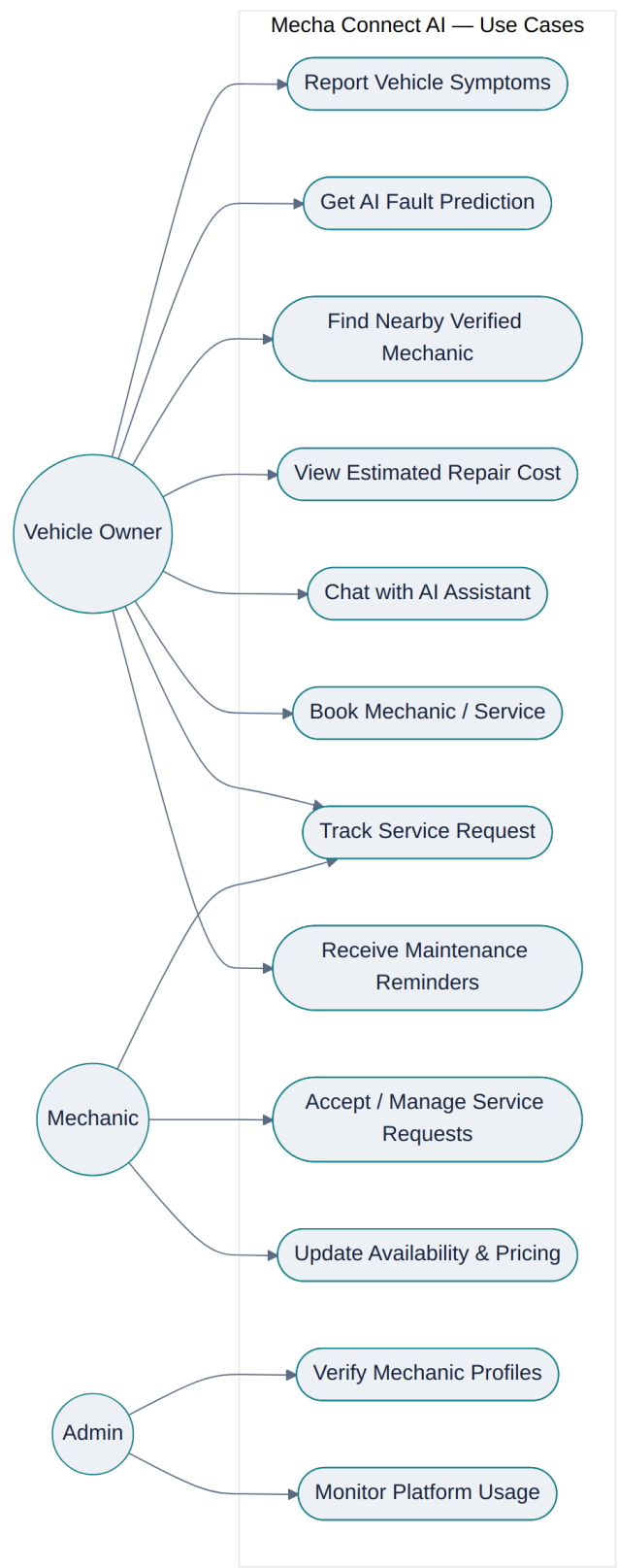
End-to-end application design — architecture, behavior, data, and interaction views

1. System Architecture Diagram



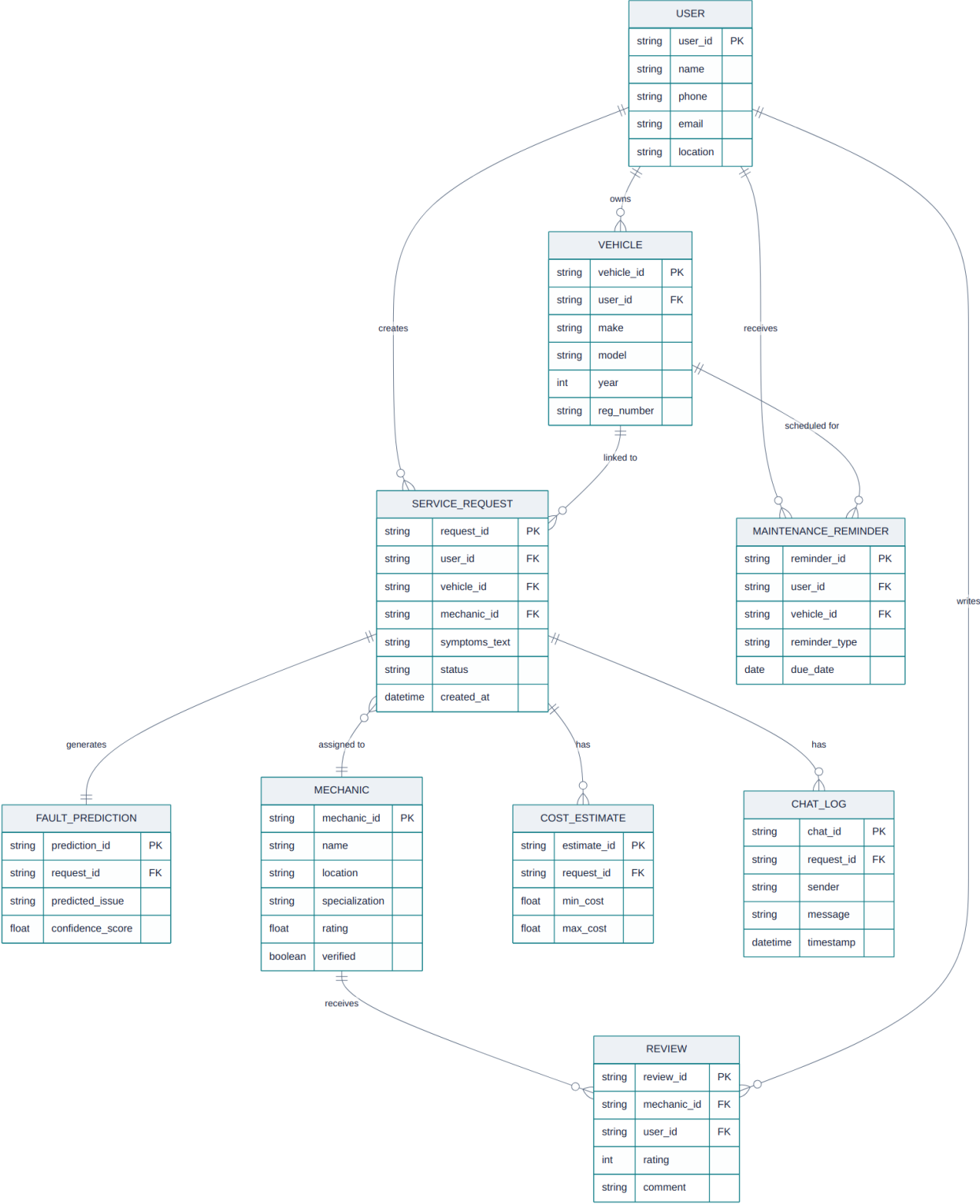
Layered view of the application: Flutter client, Python API layer, the five AI/ML services, the data layer, and external integrations (Maps, notifications).

2. Use Case Diagram



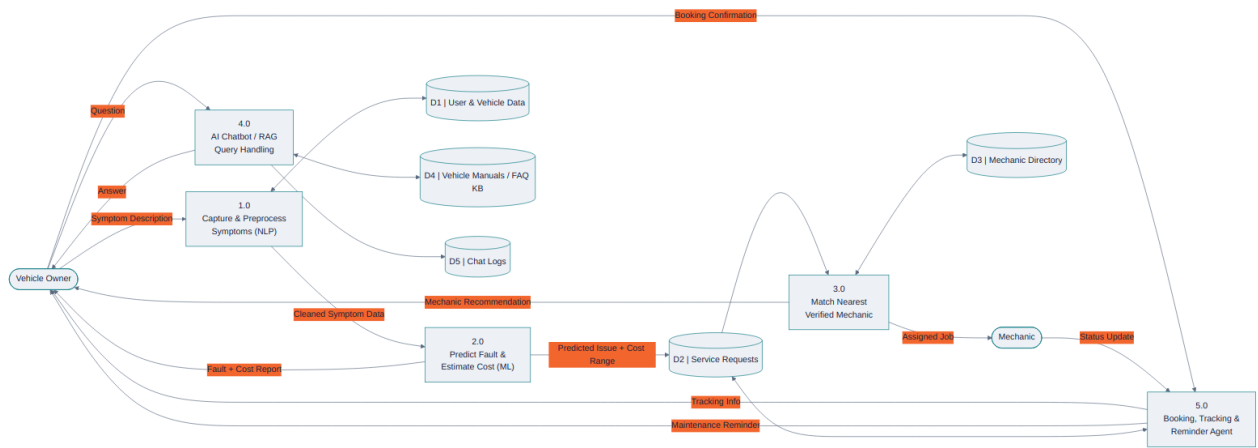
Core interactions for the three actors — Vehicle Owner, Mechanic, and Admin — across diagnosis, booking, chat, and platform-management use cases.

3. Entity-Relationship (ER) Diagram



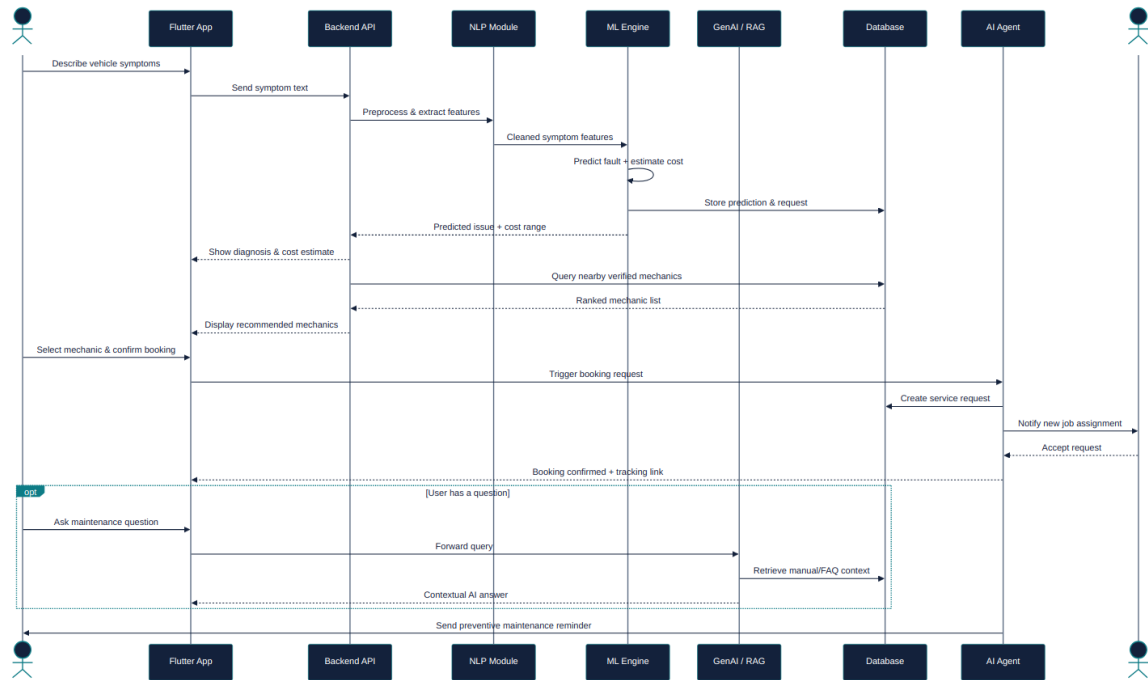
Relational schema covering users, vehicles, service requests, fault predictions, cost estimates, mechanics, reviews, chat logs, and maintenance reminders.

4. Data Flow Diagram (Level 1)



Movement of data between the Vehicle Owner, Mechanic, five core processes, and their supporting data stores.

5. Sequence Diagram — Booking Flow



Message flow from symptom submission through fault prediction, mechanic matching, booking confirmation, optional chatbot Q&A; and reminder delivery.